Hyuksoon Jang

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Education

M.S. **Pohang University of Science and Technology** — Aug 2023 – Present
Computer Science and Engineering (advised by Kyungmin Bae)

B.S. **Chungbuk National University** — Mar 2019 – Mar 2023
Computer Engineering (advised by Kyuri Jo))

Publications

Model Checking of WebAssembly Software Using Maude — 2025
Hyuksoon Jang , Kyungmin Bae. Korea Computer Congress(To appear, 2025)

Inferring transcriptomic cell states and transitions from time series transcriptome data — 2021
Jo K, Sung IY, Lee D, **Jang H**, Kim S. Scientific Reports(2021)

Projects & Experience

Model Checking of WebAssembly Software with WASM-DSL — 2024 - Present

- Developing a model-checking framework for WebAssembly software to verify distributed-system properties
- Implementing compilation rules that translate the WASM-DSL into Maude rules to automatically conform to the WebAssembly specification
- Developing an efficient state-reduction algorithm to address the state-explosion problem

MimicStudio: Web service that automatically generates 3D product models — 2022

- Led the team and implemented the neural-rendering server with Instant-NGP
- Implemented an upscaling algorithm to improve the resolution of 3D videos

COSMIC: AI-Driven Web Projects Club — 2020 - 2022

- Founded and led the club; built web projects including an AI-based security-camera service and an undergraduate course-recommendation platform
- Designed and supervised several instructional toy projects to teach web frameworks such as React and Django

TEACHING

CSED415: Computer Security (TA) — 2025 Spring

5110160: Machine Learning (TA) — 2021 Fall

0622014: Applied Computer Programming (TA) — 2020 Fall, 2021 Fall

0914002: Fundamental Computer Programming (TA) — 2021 Spring, 2022 Spring

AWARDS & FUNDING

Master's Research Fellowship — National Research Foundation of Korea — 2024–2025

Lotte Scholarship — 2020–2023

Daeshin Freight Scholarship	2020–2023
Software Maestro Program Grant, 13th Cohort — Ministry of Science and ICT	2022
Excellence Scholarship — Software-Centered University Program, CBNU	2020–2022